No ARP Spoofing:

MiniNet:

```
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.57 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.937 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.040 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.037 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.083 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.025 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.030 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.031 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.050 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.035 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.030 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.100 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.104 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.069 ms
^C
--- 10.0.0.2 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13317ms
```

Wireshark:

1	0.000000000	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2050,
2	0.000013593	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2050,
3	0.001352914	00:00:00_00:00:02	00:00:00_00:00:01	ARP	42	Who has 10.0.0.1? Tell 10.0.0.2
4	0.001360408	00:00:00_00:00:01	00:00:00_00:00:02	ARP	42	10.0.0.1 is at 00:00:00:00:00:01
5	1.023888505	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2050,
6	1.023905954	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2050,
7	2.047689519	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2050,
8	2.047707106	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2050,
9	3.073938589	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2050,
10	3.073967887	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2050,
11	4.074746700	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2050,
12	4.074766811	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2050,
13	5.121572574	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2050,
14	5.121589564	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2050,
15	6.149171311	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2050,
16	6.149225999	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2050,

ARP Spoofing without mitigation:

MiniNet:

```
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.67 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.345 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.029 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.039 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.048 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.134 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.037 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.053 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.074 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.026 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.032 ms
^C
--- 10.0.0.2 ping statistics ---
32 packets transmitted, 17 received, 46.875% packet loss, time 31853ms
```

Wireshark:

19	9.325448885	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2bba, s
20	9.325464949	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2bba, s
21	10.373266135	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2bba, s
22	10.373285146	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x2bba, s
23	10.502428362	00:00:00_00:00:03	00:00:00_00:00:01	ARP	42	10.0.0.2 is at 00:00:00:00:00:03
24	11.393367147	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2bba, s
25	12.416794525	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2bba, s
26	12.502374210	00:00:00_00:00:03	00:00:00_00:00:01	ARP	42	10.0.0.2 is at 00:00:00:00:00:03
27	13.445732809	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2bba, s
28	14.464903834	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2bba, s
29	14.504971695	00:00:00_00:00:03	00:00:00_00:00:01	ARP	42	10.0.0.2 is at 00:00:00:00:00:03
30	15.488814765	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x2bba, s
31	16.511406138	00:00:00_00:00:03	00:00:00_00:00:01	ARP	42	10.0.0.2 is at 00:00:00:00:00:03

ARP Spoofing with mitigation:

MiniNet:

```
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.30 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.11 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=1.24 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=1.24 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=2.65 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=3.20 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=1.84 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=3.18 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=1.46 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=3.45 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=2.78 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=1.44 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=1.56 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=1.29 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.999 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=2.62 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=2.62 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=2.23 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=3.66 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=4.64 ms
^C
--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19057ms
```

POX:

```
dominik@dominik-VirtualBox:~/pox$ cd ~/pox && ./pox.py misc.ipmac
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:misc.ipmac:IP-MAC detector + mitigation running
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:misc.ipmac:Set switch to forward all packets to controller
INFO:misc.ipmac:Monitoring switch: [00-00-00-00-00-01 1]
INFO:misc.ipmac:Learned: 10.0.0.3 -> 00:00:00:00:00:03 on port 3
INFO:misc.ipmac:Flooded ARP request for unknown: 10.0.0.2
INFO:misc.ipmac:Flooded ARP request for unknown: 10.0.0.1
INFO:misc.ipmac:Learned: 10.0.0.2 -> 00:00:00:00:00:02 on port 2
INFO:misc.ipmac:Flooded ARP request for unknown: 10.0.0.1
INFO:misc.ipmac:ARP proxy reply: 10.0.0.3 is at 00:00:00:00:00:03 -> port 2
INFO:misc.ipmac:Learned: 10.0.0.1 -> 00:00:00:00:00:01 on port 1
INFO:misc.ipmac:ARP proxy reply: 10.0.0.3 is at 00:00:00:00:00:03 -> port 1
INFO:misc.ipmac:ARP proxy reply: 10.0.0.2 is at 00:00:00:00:00:02 -> port 1
INFO:misc.ipmac:ARP proxy reply: 10.0.0.1 is at 00:00:00:00:00:01 -> port 2
INFO:misc.ipmac:ARP proxy reply: 10.0.0.1 is at 00:00:00:00:00:01 -> port 3
INFO:misc.ipmac:ARP proxy reply: 10.0.0.2 is at 00:00:00:00:00:02 -> port 3
WARNING:misc.ipmac:*** ARP SPOOF DETECTED — DROPPED ***
WARNING:misc.ipmac:  IP:        10.0.0.2
WARNING:misc.ipmac:  Known MAC: 00:00:00:00:00:02
WARNING:misc.ipmac:  Spoof MAC: 00:00:00:00:00:03
WARNING:misc.ipmac:*** ARP SPOOF DETECTED — DROPPED ***
WARNING:misc.ipmac:  IP:        10.0.0.2
WARNING:misc.ipmac:  Known MAC: 00:00:00:00:00:02
WARNING:misc.ipmac:  Spoof MAC: 00:00:00:00:00:03
WARNING:misc.ipmac:*** ARP SPOOF DETECTED — DROPPED ***
WARNING:misc.ipmac:  IP:        10.0.0.1
WARNING:misc.ipmac:  Known MAC: 00:00:00:00:00:01
```

MiniNet on h3 side:

```
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:2
root@dominik-VirtualBox:/home/dominik# arpspoof -i h3-eth0 -t  10.0.0.1 10.0.0.
2
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:1 0806 42: arp reply 10.0.0.2 is-at 0:0:0:0:0:3
```

```
root@dominik-VirtualBox:/home/dominik# arpspoof -i h3-eth0 -t  10.0.0.2
1
0:0:0:0:0:3 0:0:0:0:0:2 0806 42: arp reply 10.0.0.1 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:2 0806 42: arp reply 10.0.0.1 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:2 0806 42: arp reply 10.0.0.1 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:2 0806 42: arp reply 10.0.0.1 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:2 0806 42: arp reply 10.0.0.1 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:2 0806 42: arp reply 10.0.0.1 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:2 0806 42: arp reply 10.0.0.1 is-at 0:0:0:0:0:3
0:0:0:0:0:3 0:0:0:0:0:2 0806 42: arp reply 10.0.0.1 is-at 0:0:0:0:0:3
```

Wireshark:

7	0.000515171	00:00:00_00:00:01	00:00:00_00:00:02	ARP	42	10.0.0.1 is at 00:00:00:00:00:01
8	0.044581683	00:00:00_00:00:03	00:00:00_00:00:01	ARP	42	10.0.0.3 is at 00:00:00:00:00:03
9	0.044601125	00:00:00_00:00:02	00:00:00_00:00:01	ARP	42	10.0.0.2 is at 00:00:00:00:00:02
10	19.968942301	fe80::200:ff:fe00:1	ff02::2	ICMPv6	70	Router Solicitation from 00:00:00:00:00:01
11	19.970963436	fe80::200:ff:fe00:1	ff02::2	ICMPv6	70	Router Solicitation from 00:00:00:00:00:01
12	28.159783753	fe80::200:ff:fe00:3	ff02::2	ICMPv6	70	Router Solicitation from 00:00:00:00:00:03
13	28.538410840	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x4cdf, seq=1/256,
14	28.539694645	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x4cdf, seq=1/256,
15	29.539741217	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x4cdf, seq=2/512,
16	29.540834539	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x4cdf, seq=2/512,
17	30.540490899	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x4cdf, seq=3/768,
18	30.541717339	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x4cdf, seq=3/768,
19	31.542089539	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x4cdf, seq=4/1024,